

No m -cycles of the $3n-1$ map for $m \leq 61$

The Simons–de Weger template on the negative side, from a verification floor of 2^{51}

Philippe Cochin

philippe@cochin.fr

ORCID 0009-0004-1939-3382

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Preprint. One external input, Rhin’s effective measure for linear forms in $\log 2$ and $\log 3$; one computational input, the laboratory’s own verification floor, a CPU certificate to 2^{44} and a GPU sweep to 2^{51} . Everything else is proved below or is exact arithmetic recorded with its margins, and every number in the tables is recomputed by a second, independent route (Section 8). Neither the $3n-1$ cycle question nor divergence on either map is settled here.

Abstract

Let $g(y) = y/2$ for even y and $g(y) = (3y-1)/2$ for odd y , the $3n-1$ map on the positive integers, equivalently the shortcut $3n+1$ map on the negative integers. Its known cycles are 1, (5, 7, 10) and the eleven-element cycle at 17, and every start below 2^{51} reaches one of them. An m -cycle is a cycle with m maximal runs of odd elements. We transpose the template of Simons and de Weger for $3n+1$ m -cycles to this map, deriving its constants on this side rather than borrowing them: the odd step subtracts, so in the variable $u = y-1$ an odd run is exact multiplication by $3/2$, a run of a odd steps starts at $y \geq 2^a + 1$, the cycle equation bounds the linear form $\Lambda = o \log 3 - K \log 2$ by $m/(x_{\min}-1)$ with constant one, and successive local minima obey $u_{i+1} < u_i^{\log_2 3}/2$. Those three facts are three constraints on one vector, and using them together rather than separately is a fourth ingredient: a cycle cannot keep all its local minima near the floor and still carry its odd steps, because the chaining limits how fast the minima can climb. With Rhin’s bound this gives: the $3n-1$ map has no m -cycle with $1 \leq m \leq 61$ other than the two known ones. For $m \leq 2$ that is a floor-dependent form of a theorem Simons proved without any floor; for $3 \leq m \leq 61$ we know of no earlier statement. For $m \leq 52$ no admissible cycle length lies below Rhin’s ceiling; for $53 \leq m \leq 61$ the admissible lengths are excluded, the closest by 0.1 bits. At $m = 62$ one length remains, 83130157078217, with 0.3 bits of room, and a floor of $2^{55.25}$ removes it. The same tables give $m \leq 49$ from 2^{40} , $m \leq 51$ from 2^{44} , $m \leq 68$ from 2^{56} , $m \leq 74$ from 2^{60} and $m \leq 89$ from 2^{68} ; run on the $3n+1$ side at Hercher’s floor the same machinery returns $m \leq 90$ against his published 91. No published verification floor and no m -cycle theorem with $m \geq 3$ is known to us for this map.

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1. The map, the cycles, and what is known

Write g for the shortcut $3n-1$ map above. Its orbits on the positive integers are the orbits of the shortcut Collatz map $T(x) = x/2$, $(3x+1)/2$ on the negative integers under $x = -y$: $(3(-y)+1)/2 = -(3y-1)/2$. Three cycles are known,

$$(1), \quad (5, 7, 10), \quad (17, 25, 37, 55, 82, 41, 61, 91, 136, 68, 34),$$

with $(K, o) = (1, 1), (3, 2), (11, 7)$ steps and odd steps. Whether these are all the cycles, and whether every orbit is bounded, are the $3x-1$ questions of Lagarias’s bibliographies [L85], open.

The floor. Every $1 \leq y < 2^{51}$ reaches one of the three cycles. Below 2^{44} this is the laboratory's CPU computation of 19–20 September 2026, descent by strong induction on odd starts in one hundred and twelve chunks: sixteen with a plain walker through 2^{40} (greatest step count 544 against a cap of 4000), then ninety-six chunks of $5 \cdot 2^{35}$ starts with a residue sieve modulo 2^{24} and a 2^{16} -step block map as an accelerator (the sieve is exact by Lemma 5), 8246337208320 odd starts counted exactly (greatest step count 704 against a cap of 40000, counted in blocks of sixteen); the seven odd starts the first sixteen chunks had skipped, found from their own printed counts, are verified by full iteration. The range $[2^{44}, 2^{51})$ was verified on 21 September 2026 by a CUDA implementation of the same descent on one RTX 5090: the same sieve, then Barina's domain switch [B21] with the sign flipped, which is Lemma 1 below, a whole odd run in one multiplication and a whole even run in one shift, on a 128-bit state with an overflow report. It was calibrated first on the CPU's range, where it reproduces every count, both peaks and the plain walker's step counts exactly, and cross-checked against the CPU walker on three 2^{37} windows inside the new range; 1117103813820416 odd starts, no failure, no new cycle, no overflow, greatest step count 847, peak about $2^{97.0}$, in 3517 seconds. No start failed and no new cycle appeared in either range. The verifier sources, the chunk reports, the calibration and the coverage checks are archived with the branch `negative_floor_3x1`. We found no published floor for this map: the only statement in print that the searches of Section 8 reached is an unattributed comment on OEIS A037084, “up to at least 100000000”. The certificate is a computation and is labelled as one throughout.

m -cycles. A cycle with at least one even step splits into $m \geq 1$ maximal runs of odd elements alternating with runs of even elements. The first element of an odd run is a local minimum, the element after its last odd step a local maximum. The two known cycles with an even step have $m = 1$ and $m = 2$. Simons [S07] proves that g has exactly one nontrivial 2-cycle, at 17, and does so without any floor, by de Weger's bounds on $0 < 3^K - 2^{K+L} < 3^{0.89K}$, Steiner's 1-cycle theorem [St77] and an extremal inequality; he states that the floor-free route stops at $m \geq 3$. For $3n + 1$, Simons [Si05] excludes 2-cycles, Simons and de Weger [SdW] exclude m -cycles for $m \leq 68$ from the floor of November 2004, $X_0 = 301 \cdot 2^{50} > 3.3889 \cdot 10^{17}$, and Hercher [H23] reaches $m \leq 91$ from $695 \cdot 2^{60}$ with the arrangement of the valleys; Sinisalo [Si03] tabulates the cycle lengths a floor alone leaves on that side, the table this note's m -free row transposes. The template of [SdW] has five steps: a floor; a bound on Λ from the cycle equation (their Lemma 4 and Corollary 5); a chaining of the local minima (Lemmas 6, 7); Crandall's lemma generalized for a lower bound on the length (Lemma 10, Corollary 11) against Rhin's bound for an upper one (Lemmas 12, 14), sharpened through the partial quotients of $\log_2 3$ (Lemma 16); and de Weger's approximation lattice where a window remains (Lemma 18). At their floor the three stages give $m \leq 57$, $m \leq 63$ and $m \leq 68$. We know of no application of it to $3n - 1$; its first step, a verification floor, was missing. [S08] carries the template to $3x + q$ and to $px + q$, but under the standing hypothesis $q = 1$ or $q \geq 5$ prime, which excludes $q = -1$; its Section 5 treats Guy's permutation, a different map, and the paper makes no statement about $3x - 1$ anywhere. This note runs all five steps with the constants of this side. The exact enumeration of the admissible lengths by the three-gap walk is the list their approximation lattice produces ([SdW] Section 7: a two-dimensional lattice whose points of small norm are the pairs (K, L) in the window of Corollary 5 below the ceiling, found by a reduced basis and a search, each then “checked for fulfilling Corollary 5 and Lemma 7”); the two tests applied here to each length are those two. Run on the $3n + 1$ side at their floor, the machinery below returns their Lemma 18 (Section 5).

2. Notation

Throughout $x = \log 2 / \log 3$, $\delta = \log_2 3 = 1/x$. For a cycle C of g with K steps and o odd steps put

$$\Lambda(C) = o \log 3 - K \log 2.$$

For odd y put $u = y - 1$. The local minima of an m -cycle are $y_1, \dots, y_m$ in cyclic order, $u_i = y_i - 1$, the run from y_i has a_i odd steps, and $r_i \geq 1$ halvings follow it. $x_{\min}$ is the least element of C . The floor is $X_0 = 2^{51}$.

3. Six lemmas on the negative side

Lemma 1 (odd runs). Let $y \geq 3$ be odd, $u = y - 1$, $a = v_2(u)$. Then

$$g^k(y) - 1 = \left(\frac{3}{2}\right)^k (y - 1) \quad (0 \leq k \leq a),$$

$g^k(y)$ is odd for $k < a$ and $g^a(y)$ is even. Consequently a run of a odd steps starts at $y \geq 2^a + 1$, and its local maximum $M = g^a(y)$ satisfies $M \leq u^\delta + 1$.

Proof. For odd y , $g(y) - 1 = (3y - 1)/2 - 1 = \frac{3}{2}(y - 1)$. While the iterates stay odd this repeats, so $g^k(y) = 1 + 3^k u / 2^k$ as long as $g^j(y)$ is odd for $j < k$; and $1 + 3^k u / 2^k$ is an odd integer exactly when $2^{k+1} \mid u$, since 3^k is odd. Hence the first $a = v_2(u)$ iterates are odd and the next is even. Then $u \geq 2^a$ gives $y \geq 2^a + 1$, and $(3/2)^a \leq (3/2)^{\log_2 u} = u^{\log_2(3/2)} = u^{\delta-1}$ gives $M = 1 + (3/2)^a u \leq 1 + u^\delta$. $\square$

This is Hercher's Lemma 8 [H23] with $y \equiv 1 \pmod{2^a}$ in place of $y \equiv -1$: the floor $2^a + 1$ is larger by two, and it is attained at 5 ($a = 2$) and 17 ($a = 4$).

Lemma 2 (cycle equation). On a cycle C with at least one even step,

$$\Lambda(C) = - \sum_{y \in C \text{ odd}} \log\left(1 - \frac{1}{3y}\right) > 0, \quad \text{so} \quad 3^o > 2^K,$$

and $\Lambda(C) < \sum_{y \in C \text{ odd}} \frac{1}{3y - 1}$. The run from a local minimum y contributes less than $1/(y - 1)$. Hence, for an m -cycle,

$$\Lambda(C) < \sum_{i=1}^m \frac{1}{y_i - 1} \leq \frac{m}{x_{\min} - 1}.$$

Proof. An odd step multiplies by $\frac{3}{2}(1 - \frac{1}{3y})$ and an even step by $\frac{1}{2}$; around the cycle the product is 1, so $3^o 2^{-K} \prod_{\text{odd}} (1 - 1/(3y)) = 1$, which is the displayed identity, and the product is below 1. The bound $-\log(1 - t) \leq t/(1 - t)$ at $t = 1/(3y)$ gives $1/(3y - 1)$. Along the run from y , Lemma 1 gives $3g^k(y) - 1 = 3(3/2)^k u + 2 > 3(3/2)^k u$, so the run contributes less than $\frac{1}{3u} \sum_{k \geq 0} (2/3)^k = 1/u$. Finally the least element of C is odd (an even least element would halve to something smaller) and its predecessor is even (an odd predecessor y' would have $y' < (3y' - 1)/2 = x_{\min}$), so $x_{\min}$ is a local minimum and every $y_i \geq x_{\min}$. $\square$

This is [SdW] Lemma 4 and Corollary 5 with the sign flipped. The corresponding bound on the positive side, Hercher's Corollary 17, carries the constant 97/54, and [SdW] Lemma 4 has $\sum 1/x_i$ over the minima themselves. Here it is $1/(y_i - 1)$: the odd step subtracts, so the run's elements sit at or above the geometric progression rather than below it.

Lemma 3 (chaining). For consecutive local minima, $u_{i+1} < u_i^\delta/2$. Hence, with $B(m) = (\delta^m - 1)/(\delta - 1)$,

$$o = \sum_{i=1}^m a_i \leq B(m) \log_2 u_1 - \frac{B(m) - m}{\delta - 1}, \quad K < \delta o,$$

and therefore, taking $y_1 = x_{\min}$,

$$\log_2(x_{\min} - 1) > L_{\min}(K, m) := \frac{1}{B(m)} \left(\frac{K}{\delta} + \frac{B(m) - m}{\delta - 1} \right).$$

Proof. $y_{i+1} = M_i/2^{r_i}$ with $r_i \geq 1$ and $M_i \leq u_i^\delta + 1$ (Lemma 1), so $u_{i+1} = M_i/2^{r_i} - 1 \leq (u_i^\delta + 1)/2 - 1 < u_i^\delta/2$. Taking logarithms, $\log_2 u_{i+1} < \delta \log_2 u_i - 1$, and by induction $\log_2 u_i \leq$

$\delta^{i-1} \log_2 u_1 - (\delta^{i-1} - 1)/(\delta - 1)$. Summing over i and using $a_i = v_2(u_i) \leq \log_2 u_i$ gives the bound on o ; $K \log 2 < o \log 3$ is $\Lambda > 0$. Solving for $\log_2 u_1$ gives $L_{\min}$. $\square$

The chaining is tight on the 17-cycle: $u_1 = 16$, $u_2 = 40$, and $16^\delta/2 = 40.5$. It is [SdW] Lemma 6 in the variable u : their $x_{i+1} < b^\delta x_i^\delta$ with $b = (1 + X_0^{-1})/2^{1/\delta}$ becomes exact here, with no X_0 in the constant, because $u = y - 1$ is the conjugated variable itself. The bound on o is their Lemma 7 read as a lower bound on the least element rather than as an upper bound on Λ ; their constant $c_m = 2^{(m/\delta)(\delta-1)/(\delta^m-1)} b^{\delta/(\delta-1)-m/(\delta^m-1)}$ is, at $b^\delta = 1/2$, the $2^{-(B-m)/((\delta-1)B)}$ of Lemma 3.

Lemma 4 (admissible lengths). If an m -cycle has $x_{\min} \geq X_0$, then $0 < \Lambda < m/(X_0 - 1) < \log 3$, so $o = \lceil Kx \rceil$ and

$$\Lambda(K) = \log 3 (1 - \{Kx\}) < \frac{m}{X_0 - 1}.$$

Hence the admissible lengths, those a cycle above the floor can have, are exactly the K with $1 - \{Kx\} < m/((X_0 - 1) \log 3)$, and below any bound they are finite in number.

Proof. Lemma 2 gives $0 < \Lambda < m/(x_{\min} - 1) \leq m/(X_0 - 1)$, and $m/(X_0 - 1) < \log 3$ at every (m, X_0) in Section 5. Then $0 < o \log 3 - K \log 2 < \log 3$ puts o strictly between Kx and $Kx + 1$, and Kx is irrational, so $o = \lceil Kx \rceil$ and $\Lambda = \log 3 (o - Kx) = \log 3 (1 - \{Kx\})$. Finiteness below a bound is immediate. $\square$

How the finite set is listed. In practice the returns of $\{Kx\}$ into the window are walked: from the two least returns q_1, q_2 , found among the convergents and semiconvergents of x , each further member is the previous one plus q_1, q_2 or $q_1 + q_2$, which is the classical three-distance structure [3D]; the least member is Crandall's lower bound on the length. Nothing below depends on that recursion being the right one. Theorem 8 quantifies over *every* admissible K below the ceiling, and the lists used to verify it were produced twice by unrelated means, by the walk and by an exhaustive integer sieve over $K = iQ + j$ at 2^{256} scale, which agree (Section 8).

[SdW] bound the same set from both ends before listing it: Lemma 10 and Corollary 11 give the least convergent denominator the floor admits, Lemma 16 caps the length through the largest partial quotient below that index, and Section 7 lists what lies between by an approximation lattice. The enumeration below replaces all three.

Lemma 5 (contracting prefixes, and the sieve the floor uses). Let w be the first j parities of the orbit of y , with a of them odd. Then

$$2^j g^j(y) = 3^a y - D(w), \quad D(w) \geq 0,$$

with D a function of the word alone. The parity of $g^i(y)$ depends only on $y \bmod 2^{i+1}$, so w depends only on $y \bmod 2^j$. Consequently, if $3^a < 2^j$ then $g^j(y) < y$ for **every** y in that residue class, with no lower threshold on y , and a descent verification may skip the class entirely.

Proof. Induction on j . At $j = 0$, $D = 0$. An even step has $2g(y') = y'$, so $2^{j+1} g^{j+1}(y) = 2^j g^j(y)$ and D is unchanged; an odd step has $2g(y') = 3y' - 1$, so $2^{j+1} g^{j+1}(y) = 3(3^a y - D) - 2^j$ and D becomes $3D + 2^j$. Both keep $D \geq 0$ and depend on the word alone. Since 3^a is odd, $2^i g^i(y) = 3^{a_i} y - D_i$ determines $g^i(y) \bmod 2$ from $y \bmod 2^{i+1}$. Finally $g^j(y) = (3^a y - D)/2^j \leq 3^a y / 2^j < y$ when $3^a < 2^j$. $\square$

The absence of a threshold is the sign flip again: for $3n + 1$ the word constant is added, so a contracting prefix drops only the members of the class above an explicit bound, and the sieve must carry that bound. Here it is subtracted and the class falls entire. This is what makes the floor computation of Section 1 exact rather than approximate: at $j = 24$ the 286581 classes that never contract are the only ones walked, a count that is OEIS A076227(24).

Lemma 6 (valley count). Let C be an m -cycle with $x_{\min} \geq X_0$ and o odd steps, write $b_i = \log_2 u_i$ and $L_0 = \log_2(X_0 - 1)$. For $T \geq L_0$ let $R(T)$ be the largest $r \leq m$ with

$$rT + \sum_{t=1}^{m-r} \left(\delta^t T - \frac{\delta^t - 1}{\delta - 1} \right) \geq o.$$

($R(T) = 0$ if no $r \geq 0$ qualifies). Then, for every $T \geq L_0$,

$$\Lambda(C) < R(T) 2^{-L_0} + (m - R(T)) 2^{-T}.$$

Proof. Three properties of the b_i . Each $u_i \geq x_{\min} - 1 \geq X_0 - 1$, so $b_i \geq L_0$. Lemma 1 gives $u_i \geq 2^{a_i}$, so $a_i \leq b_i$ and $\sum b_i \geq \sum a_i = o$. Lemma 3 gives $u_{i+1} < u_i^\delta/2$, so $b_{i+1} < \delta b_i - 1$.

Fix $T \geq L_0$, put $S = \{i : b_i \leq T\}$ and $r = |S|$. If S is empty then every $b_i > T$, so $\Lambda < m2^{-T}$ by the last step below, and the displayed bound holds because $R(T) \geq 0$. Otherwise, for $i \notin S$ let $d_i \geq 1$ be least with $i - d_i \in S$, the indices read cyclically. Iterating $b_{j+1} < \delta b_j - 1$, and using that $z \mapsto \delta z - 1$ is increasing,

$$b_i \leq \delta^{d_i} b_{i-d_i} - \frac{\delta^{d_i} - 1}{\delta - 1} \leq \delta^{d_i} T - \frac{\delta^{d_i} - 1}{\delta - 1}.$$

The multiset $\{d_i\}$ is, on each maximal gap of length g between cyclically consecutive members of S , exactly $1, \dots, g$; the gap lengths sum to $m - r$. The right-hand side above increases with d_i , so splitting $m - r$ into several gaps gives a smaller total than one gap of length $m - r$. Hence

$$o \leq \sum_i b_i \leq rT + \sum_{t=1}^{m-r} \left(\delta^t T - \frac{\delta^t - 1}{\delta - 1} \right).$$

That right-hand side is decreasing in r , since replacing the largest ceiling term by T lowers it whenever $T \geq 1/(\delta - 1)$, which holds as $L_0 \geq 2$. So $r \leq R(T)$. Finally, by Lemma 2, $\Lambda < \sum_i 1/u_i = \sum_i 2^{-b_i} \leq r2^{-L_0} + (m - r)2^{-T}$, and that expression increases with r because $2^{-L_0} \geq 2^{-T}$. $\square$

At $T = L_0$ the lemma is exactly Lemma 2, so it is never weaker; it is stronger whenever some larger threshold does better, which is whenever o is large enough that most minima have to climb. This is the content of the arrangement behind Hercher's Main Theorem 21 [H23], which reaches the same conclusion on the $3n + 1$ side through a pigeonhole over windows of consecutive runs. Here the cyclic structure is already in Lemmas 1 to 3 and no pigeonhole is needed. Section 5 records what the two give on the same floor.

4. Rhin's ceiling

Proposition 7. Rhin's Proposition [R87, p. 160, (7)] states that for integers u_0, u_1, u_2 with $H = \max(|u_1|, |u_2|) \geq 2$,

$$|u_0 + u_1 \log 2 + u_2 \log 3| \geq H^{-13.3},$$

with no further constant. For an m -cycle of length $K \geq 2$ with an even step, $(u_0, u_1, u_2) = (0, -K, o)$ and $H = K$, so

$$2^{L_{\min}(K, m)} < x_{\min} - 1 < \frac{m}{\Lambda} \leq m K^{13.3}.$$

Let $K_3(m)$ be the least integer such that $2^{L_{\min}(K, m)} \geq 2m K^{13.3}$ for every $K \geq K_3(m)$. Then every m -cycle has $K < K_3(m)$.

Proof. Lemmas 2 and 3 with Rhin's bound. The function $K \mapsto L_{\min}(K, m) - 13.3 \log_2 K - \log_2(2m)$ is convex, negative at $K = 2$, and has a larger root; beyond it the function increases, so $K_3(m)$ is well defined, and the doubled constant covers $m/\Lambda + 1$. $\square$

Rhin's bound is the one external input of this note. It is a theorem, not a hypothesis; it is external in the sense that nothing here re-proves it and no Lean statement carries it. [SdW] Lemma 12 states it as $\Lambda > e^{-13.3(0.46057 + \log K)}$ in their odd count K , which is (7) at $H = K + L < 1.000001 \delta K$ with $0.46057 = \log \delta$; the laboratory's Paper A and its cycle-gap branch carried that form, weaker by the factor $\delta^{13.3} = 457$ when H is the length itself, and it is (7) that is used here. The general two-logarithm bound of Laurent–Mignotte–Nesterenko [LMN] would replace 13.3 by an exponent in the thousands at these heights and is not competitive; Rhin's constant is specific to $\log 2$ and $\log 3$.

5. The theorem

Theorem 8. Let $X_0 = 2^{51}$. For every $1 \leq m \leq 61$ and every length $K < K_3(m)$ with $\Lambda(K) < m/(X_0 - 1)$, at least one of

$$2^{L_{\min}(K,m)} \geq \frac{m}{\Lambda(K)}, \quad \Lambda(K) \geq \min_{T \geq L_0} \left(R(T)2^{-L_0} + (m - R(T))2^{-T} \right)$$

holds. Consequently, with Rhin’s bound, the $3n - 1$ map has no m -cycle with $1 \leq m \leq 61$ whose least element is at least 2^{51} , and by the floor no m -cycle with $m \leq 61$ other than $(5, 7, 10)$ and the cycle at 17. For $m \leq 2$ this is weaker than [S07], which needs no floor; the content of the theorem is the range $3 \leq m \leq 61$.

Proof. An m -cycle above the floor has $K < K_3(m)$ by Proposition 7, its length is admissible by Lemma 4, and its least element satisfies $2^{L_{\min}(K,m)} < x_{\min} - 1 \leq m/\Lambda(K)$ by Lemmas 2 and 3, contradicting the first display; and $\Lambda(K)$ is below the valley cap by Lemma 6, contradicting the second. The alternative is a finite exact computation: for each m the admissible lengths below $K_3(m)$ are the finitely many K of Lemma 4, listed by two unrelated methods that agree, and each is tested at eighty digits against both displays. For $m \leq 52$ there is no admissible length below $K_3(m)$ at all, so the conclusion there rests on Lemmas 2 and 4 and Proposition 7 alone. For $53 \leq m \leq 61$ the admissible lengths exist and fail the displays by the margins of Table 1. $\square$

Table 1. Rows at the floor 2^{51} . “Margin” is the largest, over the admissible $K < K_3(m)$, of the better of the two displays of Theorem 8 measured in bits; it is negative exactly when every admissible length is excluded.

m	$m/(X_0 - 1)$	$K_3(m)$	admissible $K < K_3$	least admissible	margin (bits)
1	$4.4 \cdot 10^{-16}$	155	0	—	—
2	$8.9 \cdot 10^{-16}$	495	0	—	—
5	$2.2 \cdot 10^{-15}$	3926	0	—	—
10	$4.4 \cdot 10^{-15}$	57129	0	—	—
20	$8.9 \cdot 10^{-15}$	8393107	0	—	—
30	$1.3 \cdot 10^{-14}$	1094917645	0	—	—
40	$1.8 \cdot 10^{-14}$	134716197551	0	—	—
50	$2.2 \cdot 10^{-14}$	15974496522785	0	—	—
53	$2.4 \cdot 10^{-14}$	66574098182843	1	64789416887513	−549.2
54	$2.4 \cdot 10^{-14}$	107084917386112	2	64789416887513	−328.1
55	$2.4 \cdot 10^{-14}$	172208666666214	4	64789416887513	−188.5
56	$2.5 \cdot 10^{-14}$	276877898691766	6	64789416887513	−100.4
57	$2.5 \cdot 10^{-14}$	445072680198484	10	64789416887513	−44.8
58	$2.6 \cdot 10^{-14}$	715295665905327	16	64789416887513	−9.8
59	$2.6 \cdot 10^{-14}$	1149356600587428	27	64789416887513	−1.7
60	$2.7 \cdot 10^{-14}$	1846464505724577	44	64789416887513	−0.7
61	$2.7 \cdot 10^{-14}$	2965831161057260	73	64789416887513	−0.1
62	$2.8 \cdot 10^{-14}$	4762921235638509	120	46448676696809	+0.3

The exclusion tightens steadily and ends on a knife edge: at $m = 61$ the closest admissible length clears by 0.1 bits, and it is $K = 83130157078217$, the length that survives one row later. From $m = 53$ to $m = 58$ the closest length is instead 64789416887513, the least admissible at this floor. Any weakening of the constants, in Rhin’s form, in the chaining or in the valley count, by more than 0.1 bits would move the theorem back to $m \leq 60$ or further. At $m = 62$ one length survives, 83130157078217, with 0.3 bits of room; a floor of $2^{55.25}$ removes it (Lemma 2’s $x_{\min} - 1 < m/\Lambda$), which is why 2^{56} is the next floor worth running.

Table 2. What each floor buys, by the same tables.

floor	m excluded through	first open m	lengths left at the first open m
2^{40}	49	50	103768467013, 1193652440098
2^{44}	51	52	1193652440098
2^{48}	58	59	83130157078217
2^{49}	58	59	83130157078217
2^{50}	59	60	83130157078217
2^{51}	61	62	83130157078217
2^{56}	68	69	9881527843552324
$301 \cdot 2^{50}$ ([SdW]'s floor)	69	70	9881527843552324
2^{60}	74	75	9881527843552324
2^{68}	89	90	79641170620168673833

Three comparisons calibrate the method, all of them on the other sign and so plausibility checks rather than reproductions.

The first is the sharpest. Run on the $3n + 1$ side, with the window on the contracting side of x and the same six lemmas, the machinery at Hercher's floor $695 \cdot 2^{60}$ excludes $m \leq 90$, against the $m \leq 91$ of his Main Theorem 23 [H23]. One value short of a published result obtained by the same idea is the strongest evidence available here that the transposition is faithful and that the valley count is doing the work it should.

The second is the older state of the art. At [SdW]'s floor $301 \cdot 2^{50}$ the machinery gives at least 69 where their paper gives 68. That is not an improvement on their work: their 2005 argument predates the valley arrangement entirely, and the right comparison for a paper that uses it is Hercher's. Without Lemma 6 the same floor gives 63, which is their Lemma 17 exactly, and Section 5 of the previous version of this note recorded that agreement.

The third is the reproduction of [SdW] Lemma 18 itself, unchanged from the previous version and independent of Lemma 6: at their floor the enumeration with the two tests of their Corollary 5 and Lemma 7 returns no length for $64 \leq m \leq 68$, and at $69 \leq m \leq 72$ their five pairs (K, L) at the nine places of their table, with the floors that remove them agreeing to rounding, 576.2, 584.6, 592.9, 601.3, 623.6, 632.4, 308.2, 666.8 and 705.3 times 2^{50} against their 577, 585, 593, 602, 624, 633, 309, 667 and 706. The same run lists one point more at $m = 72$, the double of their second pair, with 0.9 bits of room; their table does not carry it, and the text does not say whether non-primitive pairs are discarded.

The floor enters only through the admissibility threshold, and in steps: 2^{52} to 2^{55} buy nothing over 2^{51} , and 2^{56} buys seven values at once. Lemma 6 is worth between two and seven values of m at every floor in the table; without it the same rows read 44, 49, 49, 54, 56, 58, 63, 63, 68 and 82.

6. The cycles that exist pass the same test

The inequalities were checked where they must not exclude. With the floor set at 17 and $m = 2$, the admissible lengths include 11 and it survives the display; with the floor at 5 and $m = 1$, length 3 survives; with the floor at 18, length 11 no longer appears, since the 17-cycle's least element is below it. The cycle equation of Lemma 2 is verified in exact rationals on all three cycles, and Lemma 1's run length and identity on every odd $y < 4000$.

7. What this note does not do

It excludes no Juggler cycle. The Juggler's cycle words are, letter for letter, the words of the negative Collatz cycles ([A], Section 5.9), but a word shape does not transport a realization, so a $3n - 1$ m -cycle theorem constrains Juggler cycle words and nothing more. It does not settle the

$3n - 1$ cycle question, m being bounded, and it says nothing about divergence on either map. The lattice of admissible periods that Eliahou [E93] builds on the positive side is not transposed.

Lemmas 1 and 3 are machine-checked (`Problems.Collatz.NegativeMCycles`, twenty theorems, Lean 4 with Mathlib, no `sorry` and nothing off the kernel). The formalization writes the start of an odd run as $y = 2^a m + 1$ with m odd, which removes every truncated subtraction and puts the run in closed form, $g^k(2^a m + 1) = 3^k 2^{a-k} m + 1$ for $k \leq a$; Lemma 1’s four clauses and Lemma 3’s two halves are read off it, and the known cycles are checked against the same definitions inside the kernel. Formalizing it sharpened one hypothesis: the closed form needs no parity assumption on m , since the powers of two alone keep the state odd through the run, and oddness of m is used only to know that the run stops at a . Lemma 2 is not in Lean: its content is the cycle equation and the bound on Λ , which are real-analytic. Its first clause, $\Lambda > 0$, is the negative-cycle expansion already in Lean on the conjugate side (`neg_cycle_expanding`), together with the negative-side finance and word shape (`neg_cycle_finance`, `neg_prefix_noncontracting`); the conjugation $x = -y$ that carries one to the other is stated in this note and is not itself formalized.

8. Verification

`python -m research.juggler_sequence.negative_m_cycles` recomputes both tables in a few seconds and writes `data/research/juggler/negative_m_cycles/summary.json`; `tests/research/juggler_sequence/test_negative_m_cycles.py` checks Lemma 1 on every odd $y < 4000$, the cycle equation on the three cycles, the tightness of Lemmas 1 and 3 at 17, that the known cycles pass, that K_3 is a ceiling, that the three-gap walk agrees with a direct scan, that the archived tables reproduce, and that the same machinery with the window on the contracting side returns [SdW] Lemma 18 at their floor. The floor’s certificate is with the branch `negative_floor_3x1`: the CPU chunk reports below 2^{44} , and above it the GPU sweep’s chunk reports, its calibration on the CPU’s range and the three spot-check windows (`python -m research.juggler_sequence.negative_floor_gpu calibrate, sweep 44 51, spot`). `lake build Problems.Collatz.NegativeMCycles` checks Lemmas 1 and 3; the module is in the `Problems` barrel, so the default build covers it, and `tests/research/juggler_sequence/test_negative_m_cycles_lean.py` holds it to no `sorry`, no `native_decide` and the three standard axioms. Separately, a FLINT/Arb audit through the laboratory’s interval-arithmetic service (`python tools/check_papers_arb.py --paper D, report data/research/collatz/arb_paper_audit/d.json`) re-derives the ceilings and an interval-certified exclusion witness for every admissible length through $m = 61$, with no difference from Table 1; it takes the floor and Rhin’s theorem as inputs and is not among the fourteen pinned files.

```
Repository: https://github.com/sneakyweasel/btlab
Commit:    efaa92cbff6c3da44bb351055b32e23196de556b
Lean:      leanprover/lean4:v4.33.1
Mathlib:   v4.33.1 (lake-manifest rev 0df444a360eaa60ab8c11dca51a86af692955474)
Build:     lake build Problems.Collatz.NegativeMCycles (from formal/)
Tables:    python -m research.juggler_sequence.negative_m_cycles
Floor:     python -m research.juggler_sequence.negative_floor_gpu sweep 44 51
Recheck:   python tools/check_3n_minus_1_note_numeric.py
```

The commit is the repository state that produced the tables, the floor records and the Lean module. All fifteen files those commands read or write – the two probes, the recheck, the common paper builder with this paper’s settings, template and layout filter, the four floor records, the CUDA source, the two table files and the Lean module – are byte-identical there to the versions this paper reports; the check is mechanical and is part of the release gate. Only this text, its typeset form and the manifest of digests move afterwards, in the editorial commit that writes the line above. Version 1.0.0 of this paper was built at commit `d6e8a4ae20a5c82f46fc5e673a1bf8d1a344e29c`, before Lemma 6.

Acknowledgments and use of AI

Large language models assisted the development of this work, including the prose, proposed proof arguments, the Lean formalization, the CUDA verifier, and the computations. The two external sources were read directly and their constants taken from them rather than from memory. AI assistance is not independent mathematical validation. The author is responsible for the statements, the proofs, the code, and the decision to make this version public.

Availability and version

This is version 1.1.2 of Paper D, of 25 September 2026, at the floor 2^{51} . It is a preprint and has not been refereed. It is built by the laboratory’s common paper builder, so Section 8 names a new commit and the build command; the text is otherwise that of version 1.1.1 of the same day (doi:10.5281/zenodo.22954088), which corrected wording and precision in version 1.1.0 of 21 September 2026: two claims of novelty are hedged, Lemma 6 states its value of $R(T)$ when no r qualifies, and Section 8 records the interval-arithmetic audit; the theorem, the tables and the pinned inputs are unchanged. Version 1.0.0, deposited the same day (doi:10.5281/zenodo.22876190), proved the same theorem for $m \leq 58$; this version adds Lemma 6 and reaches $m \leq 61$ at the same floor, with the tables and the calibration regenerated, in version 1.1.0. The concept DOI 10.5281/zenodo.22876189 resolves to the latest version; the record of version 1.1.1 is at <https://zenodo.org/records/22954088>. The author’s ORCID is 0009-0004-1939-3382. The manuscript, the probe that computes the tables, the independent check that recomputes them, the Lean module, the CUDA verifier and the floor certificate with its chunk reports are in the repository sneakyweasel/btlab at the commit named in Section 8; `python tools/build_paper.py D` rebuilds this document from its canonical Markdown and `python tools/build_paper.py D --check` verifies every generated copy against the manifest. The verification floor is a computation of this laboratory, not a result from the literature, and is labelled as one throughout; a higher floor raises the theorem’s m and would be a new version rather than a correction of this one.

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